Toward a Brenier Theorem for Power-Distance Gromov–Wasserstein Transport

Self-consistent costs, quotient lifting, and finite transport branches

Gabriel Peyré

July 2026

Abstract

Let $0 < p \leq 2$, and consider Gromov–Wasserstein transport with distortion $(\|x - x'\|^p - \|y - y'\|^p)^2$. Schoenberg's theorem realizes each power distance as a squared distance in a Hilbert feature space. After centering these features, the GW objective depends on a coupling through one scalar mixed moment and one Hilbert–Schmidt cross-covariance operator. This yields an exact representation as ordinary optimal transport with a learned cross-space cost, regularized by the squared norm of a tensor-product RKHS. Every GW minimizer therefore solves a linear OT problem for its self-consistent learned cost.

This reduction reveals two distinct routes to a Brenier-type result. First, an m-twisted self-consistent cost forces every conditional plan to have at most m atoms, hence represents a prescribed GW optimizer as a measurable mixture of at most m maps. Second, conditional concavity gives a replacement principle: every optimizer of the frozen linear problem remains GW-optimal. A twisted transport problem can then be solved on a quotient and lifted through nonatomic source fibers, producing a genuine Monge optimizer even when the original twist fibers are infinite. In particular, if the learned operator has rank R, absolute continuity of its $(R + 1)$-dimensional quotient and atomlessness along its fibers imply existence of a GW-optimal map; a coarea criterion makes these hypotheses checkable.

We also give differential, multijet-transversality, and analytic-definable criteria for finite twist. Analyticity alone is shown to be insufficient. For $p = 2$, the feature spaces are finite dimensional and both mechanisms can be analyzed sharply. If the source lies in $\mathbb{R}^n$, the target lies in $\mathbb{R}^d$, $n > d$, the source has a density, and an optimal cross-covariance has full row rank, then that optimizer is induced by a map. The same conclusion holds without a strict dimension gap when the target lies on a centered sphere. In equal dimensions and without this extra geometry, the learned cost is only two-twisted: its gradient can have two target preimages. This recovers the mechanism behind the known optimal two-map theorem, while quotient lifting recovers the low-rank Monge theorem and yields an additional map regime under a strict source dimension gap. The note separates proved statements, known results, and the remaining open step for $0 < p < 2$.

1 Problem, scope, and known results

The goal is to understand when the relaxation from deterministic correspondences to couplings is exact *at the level of minimizers*. The distinction matters: graph couplings can approximate general couplings when the source is nonatomic, but the approximating maps need not converge to a map.

1.1 Power-distance GW and its Monge restriction

Let $\mathcal{X} \subset \mathbb{R}^n$ and $\mathcal{Y} \subset \mathbb{R}^d$ be compact, let $\alpha \in \mathcal{P}(\mathcal{X})$ and $\beta \in \mathcal{P}(\mathcal{Y})$, and fix $0 < p \leq 2$. Write

$$d_{p,\mathcal{X}}(x, x') = \|x - x'\|^p , \qquad d_{p,\mathcal{Y}}(y, y') = \|y - y'\|^p .$$

The inner exponent p should not be confused with the outer quadratic distortion, which remains fixed throughout this note. Thus $p = 1$ is the usual quadratic GW problem built from Euclidean distances, whereas $p = 2$ is the squared-Euclidean-dissimilarity model studied in Dumont et al. (2025).

Definition 1.1 (Power-distance GW problem)**.** For $\pi \in \Pi(\alpha, \beta)$, set

$$\mathcal{E}_p(\pi) := \iint \left(d_{p,\mathcal{X}}(x, x') - d_{p,\mathcal{Y}}(y, y')\right)^2 \mathrm{d}\pi(x, y)\, \mathrm{d}\pi(x', y'). \tag{1}$$

The Kantorovich and Monge values are

$$\mathrm{GW}^2_{p,2}(\alpha, \beta) := \min_{\pi \in \Pi(\alpha,\beta)} \mathcal{E}_p(\pi), \tag{2}$$

$$\mathrm{GM}^2_{p,2}(\alpha, \beta) := \inf_{T_\sharp \alpha = \beta} \mathcal{E}_p\left((\mathrm{Id}, T)_\sharp \alpha\right), \tag{3}$$

with value $+\infty$ in (3) if no admissible map exists.

Compactness makes existence on the Kantorovich side immediate.

Proposition 1.2 (Existence of a coupling minimizer)**.** *The minimum in* (2) *is attained.*

Proof. The set $\Pi(\alpha, \beta)$ is narrowly compact. The integrand in (1) is continuous and bounded on $(\mathcal{X} \times \mathcal{Y})^2$. Hence $\pi \mapsto \mathcal{E}_p(\pi)$ is narrowly continuous. □

The central question is whether some minimizer in (2) has the form $(\mathrm{Id}, T)_\sharp \alpha$. This is stronger than equality of the two infimum values. For nonatomic spaces, Mémoli and Needham (2024) establish broad equality results between Gromov–Monge and Gromov–Wasserstein values, but these do not imply that the coupling minimum is attained by a graph.

1.2 What is known at $p = 2$

Classical Brenier theory concerns a *linear* transport objective with quadratic ground cost: if the source has a density, the unique optimal coupling is induced by the gradient of a convex function (Brenier, 1991; Villani, 2009; Santambrogio, 2015). GW is instead quadratic in the unknown coupling, so Brenier's theorem cannot be applied before one has identified an appropriate frozen linear cost.

For GW built from inner products, this program succeeds particularly well. Vayer proved a transformed-Brenier theorem under a full-rank assumption (Vayer, 2020, Theorem 4.2.3), and Dumont et al. (2025) subsequently established existence of a Monge optimizer without that rank assumption. Squared Euclidean dissimilarities behave differently. In the notation of definition 1.1, this is exactly the case $p = 2$.

Definition 1.3 (Finite-map coupling)**.** A coupling $\pi \in \Pi(\alpha, \beta)$ is a *k-map* if there are measurable maps $T_1, \ldots, T_k : \mathcal{X} \to \mathcal{Y}$ and measurable weights $\lambda_j : \mathcal{X} \to [0, 1]$, with $\sum_{j=1}^k \lambda_j = 1$ α-almost everywhere, such that

$$\pi = \sum_{j=1}^{k} (\mathrm{Id}, T_j)_\sharp (\lambda_j \alpha).$$

Equivalently, the conditional measure π_x is supported on at most k points for α-almost every x. A *two-map* is the case $k = 2$.

The following is the current general structural theorem for squared-distance GW.

Theorem 1.4 (Known squared-distance structure Dumont et al., 2025). *Let $n \geq d$, let α and β have compact support, and assume $\alpha \ll \mathrm{Leb}^n$. For $p = 2$, an optimal two-map exists. More precisely, if*

$$S_\pi := \int (y - m_\beta)(x - m_\alpha)^\top \,\mathrm{d}\pi(x, y) \in \mathbb{R}^{d\times n}$$

is the cross-covariance of an optimal coupling, their theorem gives a single-map optimizer when $\operatorname{rank} S_\pi \leq d - 2$, *and a two-map in the remaining rank regimes. In the full-rank regime the two-map can also be described as a map/anti-map.*

The same work gives one-dimensional numerical examples in which the optimum appears genuinely two-valued. This is evidence, rather than a theorem of nonexistence, but it makes a universal single-map claim mathematically unsafe. Our objective below is therefore twofold: derive the strongest general theorem furnished by the RKHS reduction, and identify concrete hypotheses under which either a frozen optimizer has finite twist or a twisted quotient problem can be lifted to a Monge optimizer.

2 Power distances as Hilbert features

The restriction $0 < p \leq 2$ is exactly the range in which Euclidean power distances are of negative type. This converts the nonlinear-looking distance product in GW into a squared Hilbert–Schmidt norm.

2.1 Schoenberg embedding and centered kernels

Proposition 2.1 (Schoenberg feature representation). *For every $0 < p \leq 2$ and every Euclidean dimension m, there are a real Hilbert space $\mathcal{H}_{p,m}$ and an injective continuous map $\Phi_{p,m} : \mathbb{R}^m \longrightarrow \mathcal{H}_{p,m}$ such that*

$$\|x - x'\|^p = \left\|\Phi_{p,m}(x) - \Phi_{p,m}(x')\right\|^2_{\mathcal{H}_{p,m}}. \tag{4}$$

For $p = 2$, one can take $\mathcal{H}_{2,m} = \mathbb{R}^m$ and $\Phi_{2,m} = \mathrm{Id}$. For $0 < p < 2$, one possible realization is the real form of

$$\Phi_{p,m}(x)(\omega) = c_{m,p}^{1/2}\left(e^{\mathrm{i}\langle\omega,x\rangle} - 1\right) \quad in \quad L^2\left(\mathbb{R}^m, \frac{\mathrm{d}\omega}{\|\omega\|^{m+p}}\right), \tag{5}$$

for a normalization constant $c_{m,p} > 0$.

Proof. Schoenberg's theorem says that $(x, x') \mapsto \|x - x'\|^p$ is conditionally negative definite precisely in the range $0 < p \leq 2$; the standard Hilbert embedding of a negative-type semimetric gives (4) (Schoenberg, 1938). For $0 < p < 2$, the identity

$$\|x - x'\|^p = c_{m,p} \int_{\mathbb{R}^m} \frac{\left|e^{\mathrm{i}\langle\omega,x\rangle} - e^{\mathrm{i}\langle\omega,x'\rangle}\right|^2}{\|\omega\|^{m+p}} \,\mathrm{d}\omega$$

gives (5). The integral is finite near the origin because $p < 2$, and at infinity because $p > 0$. The identity map proves the endpoint $p = 2$. □

Apply proposition 2.1 separately to $\mathcal{X}$ and $\mathcal{Y}$, writing the feature maps as Φ and Ψ. Center them with respect to the marginals:

$$u(x) := \Phi(x) - \int \Phi \,\mathrm{d}\alpha, \qquad v(y) := \Psi(y) - \int \Psi \,\mathrm{d}\beta. \tag{6}$$

Let $\mathcal{H}_\alpha$ and $\mathcal{H}_\beta$ be the closed linear spans of $u(\mathcal{X})$ and $v(\mathcal{Y})$. Their reproducing kernels are

$$k_\alpha(x, x') := \langle u(x), u(x')\rangle_{\mathcal{H}_\alpha}, \qquad k_\beta(y, y') := \langle v(y), v(y')\rangle_{\mathcal{H}_\beta}. \tag{7}$$

These kernels can be evaluated without constructing the feature spaces. Set

$$r_\alpha(x) := \int d_{p,\mathcal{X}}(x,z)\,\mathrm{d}\alpha(z), \qquad D_\alpha := \iint d_{p,\mathcal{X}}(x,z)\,\mathrm{d}\alpha(x)\,\mathrm{d}\alpha(z), \tag{8}$$

$$r_\beta(y) := \int d_{p,\mathcal{Y}}(y,z)\,\mathrm{d}\beta(z), \qquad D_\beta := \iint d_{p,\mathcal{Y}}(y,z)\,\mathrm{d}\beta(y)\,\mathrm{d}\beta(z). \tag{9}$$

Double centering gives

$$k_\alpha(x,x') = \frac{1}{2}\big(r_\alpha(x) + r_\alpha(x') - d_{p,\mathcal{X}}(x,x') - D_\alpha\big), \tag{10}$$

$$k_\beta(y,y') = \frac{1}{2}\big(r_\beta(y) + r_\beta(y') - d_{p,\mathcal{Y}}(y,y') - D_\beta\big). \tag{11}$$

In particular, define

$$a(x) := k_\alpha(x,x) = \|u(x)\|^2, \qquad b(y) := k_\beta(y,y) = \|v(y)\|^2. \tag{12}$$

Their means are $\bar{a} = D_\alpha/2$ and $\bar{b} = D_\beta/2$.

2.2 The cross-covariance decomposition

For $\pi \in \Pi(\alpha,\beta)$, define the Hilbert–Schmidt operator

$$C_\pi := \int v(y) \otimes u(x)\,\mathrm{d}\pi(x,y) \in \mathrm{HS}(\mathcal{H}_\alpha, \mathcal{H}_\beta), \tag{13}$$

where $(v \otimes u)h = \langle u, h\rangle\, v$. Compactness makes this Bochner integral well defined.

Proposition 2.2 (Distance-product decomposition). *For every $\pi \in \Pi(\alpha,\beta)$,*

$$\iint d_{p,\mathcal{X}}(x,x') d_{p,\mathcal{Y}}(y,y')\,\mathrm{d}\pi(x,y)\,\mathrm{d}\pi(x',y') = 2\int a(x)b(y)\,\mathrm{d}\pi(x,y) + 2\bar{a}\bar{b} + 4\,\|C_\pi\|^2_{\mathrm{HS}}. \tag{14}$$

Consequently,

$$\mathcal{E}_p(\pi) = \mathsf{C}_p(\alpha,\beta) - 4\int a(x)b(y)\,\mathrm{d}\pi(x,y) - 8\,\|C_\pi\|^2_{\mathrm{HS}}, \tag{15}$$

where the marginal-only constant is

$$\mathsf{C}_p(\alpha,\beta) := \iint d_{p,\mathcal{X}}(x,x')^2\,\mathrm{d}\alpha(x)\,\mathrm{d}\alpha(x') + \iint d_{p,\mathcal{Y}}(y,y')^2\,\mathrm{d}\beta(y)\,\mathrm{d}\beta(y') - 4\bar{a}\bar{b}. \tag{16}$$

In particular, $\pi \mapsto \mathcal{E}_p(\pi)$ is concave on $\Pi(\alpha,\beta)$.

Proof. Let (X,Y) and (X',Y') be independent with law π. By (4) and (6),

$$d_{p,\mathcal{X}}(X,X') = a(X) + a(X') - 2\,\langle u(X), u(X')\rangle\,,$$

and similarly on $\mathcal{Y}$. Expand the product. The product of the two radial sums contributes $2\mathbb{E}[a(X)b(Y)] + 2\bar{a}\bar{b}$. Every mixed term containing one inner product vanishes because $\mathbb{E}[u(X)] = 0$ and $\mathbb{E}[v(Y)] = 0$. Finally,

$$\mathbb{E}[\langle u(X), u(X')\rangle\,\langle v(Y), v(Y')\rangle] = \langle C_\pi, C_\pi\rangle_{\mathrm{HS}} = \|C_\pi\|^2_{\mathrm{HS}}.$$

This proves (14). Expanding the square in (1) then gives (15). Its last term is the negative of the square of a linear function of π, while the preceding coupling term is linear; hence the objective is concave. □

Corollary 2.3 (Discrete permutation optimizer). *Suppose*

$$\alpha = N^{-1} \sum_{i=1}^{N} \delta_{x_i}, \qquad \beta = N^{-1} \sum_{j=1}^{N} \delta_{y_j}.$$

For every $0 < p \leq 2$*, the problem* (2) *has an optimal permutation coupling*

$$\pi_\sigma = \frac{1}{N} \sum_{i=1}^{N} \delta_{(x_i, y_{\sigma(i)})} \tag{17}$$

for some permutation σ *of* $\{1, \ldots, N\}$*.*

Proof. The coupling set is the scaled Birkhoff polytope. Represent any minimizer as a convex combination of its extreme points. Concavity and minimality imply that every extreme point carrying positive weight is also a minimizer. The Birkhoff–von Neumann theorem identifies these extreme points with the permutation couplings (17). □

This is a genuine discrete Monge statement, but its proof does not pass to continuous measures: extreme couplings over nonatomic spaces need not be graphs, and the class of graph couplings is not narrowly closed. Concavity is therefore useful structurally without making the continuous minimization easy. The feature decomposition becomes more informative after dualizing the squared Hilbert–Schmidt norm.

3 An exact RKHS-regularized learned-cost problem

The tensor-product space $\mathcal{H}_\beta \widehat{\otimes} \mathcal{H}_\alpha$ is canonically identified with $\mathrm{HS}(\mathcal{H}_\alpha, \mathcal{H}_\beta)$. Under this identification, an operator A defines the cross-space function

$$h_A(x, y) := \langle Au(x), v(y) \rangle_{\mathcal{H}_\beta}. \tag{18}$$

It belongs to the RKHS with product kernel

$$K_{\alpha,\beta}\big((x, y), (x', y')\big) = k_\alpha(x, x') k_\beta(y, y'), \qquad \|h_A\|_{K_{\alpha,\beta}} = \|A\|_{\mathrm{HS}}. \tag{19}$$

The equality of norms uses the definition of $\mathcal{H}_\alpha$ and $\mathcal{H}_\beta$ as the closed feature spans.

For a continuous cost c on $\mathcal{X} \times \mathcal{Y}$, write

$$\mathcal{L}_c(\alpha, \beta) := \min_{\pi \in \Pi(\alpha,\beta)} \int c(x, y) \, \mathrm{d}\pi(x, y). \tag{20}$$

Theorem 3.1 (Exact RKHS cost learning). *For* $A \in \mathrm{HS}(\mathcal{H}_\alpha, \mathcal{H}_\beta)$*, define*

$$c_A(x, y) := -4a(x)b(y) - 4 \langle Au(x), v(y) \rangle. \tag{21}$$

Then, for every coupling π*,*

$$\mathcal{E}_p(\pi) = \mathsf{C}_p(\alpha, \beta) + \min_A \left\{ \frac{1}{2} \|A\|_{\mathrm{HS}}^2 + \int c_A \, \mathrm{d}\pi \right\}, \qquad A_\pi = 4C_\pi. \tag{22}$$

Consequently,

$$\mathrm{GW}_{p,2}^2(\alpha, \beta) = \mathsf{C}_p(\alpha, \beta) + \min_{A \in \mathrm{HS}(\mathcal{H}_\alpha, \mathcal{H}_\beta)} \left\{ \frac{1}{2} \|A\|_{\mathrm{HS}}^2 + \mathcal{L}_{c_A}(\alpha, \beta) \right\}. \tag{23}$$

Equivalently, with $r_A(x,y) = 4a(x)b(y) + 4\langle Au(x), v(y)\rangle$,

$$\mathrm{GW}_{p,2}^2(\alpha,\beta) = \mathsf{C}_p(\alpha,\beta) - \max_{\substack{A\in\mathrm{HS}(\mathcal{H}_\alpha,\mathcal{H}_\beta)\\ \pi\in\Pi(\alpha,\beta)}} \left\{ \int r_A \,\mathrm{d}\pi - \frac{1}{2}\|A\|_{\mathrm{HS}}^2 \right\}. \tag{24}$$

The maximand is strictly concave in A for fixed π and linear in π for fixed A; it is not asserted to be jointly concave.

Proof. The Hilbert–Schmidt pairing satisfies

$$\langle A, C_\pi\rangle_{\mathrm{HS}} = \int \langle Au(x), v(y)\rangle \,\mathrm{d}\pi(x,y).$$

Completing the square gives

$$\min_A \left\{ \frac{1}{2}\|A\|_{\mathrm{HS}}^2 - 4\langle A, C_\pi\rangle_{\mathrm{HS}} \right\} = -8\|C_\pi\|_{\mathrm{HS}}^2, \qquad A_\pi = 4C_\pi.$$

Together with (15), this proves (22). Taking the joint minimum in (π, A) and minimizing first in π proves (23). Changing signs and using $c_A = -r_A$ gives (24). □

This is the promised RKHS regularizer. The fixed radial product $-4a(x)b(y)$ is not learned; the learned component lies in the tensor-product RKHS and is penalized by its squared norm. The representation is closely related to cost-regularized cross-space transport (Sebbouh et al., 2024) and to negative-type biconvex GW relaxations (Konno, 1976; Séjourné et al., 2021).

Remark 3.2 (A bilinear cost on augmented features). Introduce

$$\widehat{u}(x) := (a(x), u(x)) \in \mathbb{R}\oplus\mathcal{H}_\alpha, \qquad \widehat{v}(y) := (b(y), v(y)) \in \mathbb{R}\oplus\mathcal{H}_\beta,$$

and the block operator $\widehat{A}(s,h) := (s, Ah)$. Then the learned cost is the bilinear expression

$$c_A(x,y) = -4\left\langle \widehat{A}\widehat{u}(x), \widehat{v}(y)\right\rangle. \tag{25}$$

Only the block A is optimized and regularized; the scalar block is fixed. Thus the reduction is bilinear after a nonlinear feature lift, not in the original variables. For $p = 2$, these augmented features are $(\|x - m_\alpha\|^2, x - m_\alpha)$ and $(\|y - m_\beta\|^2, y - m_\beta)$, which lie on paraboloids. The radial coordinate of the target paraboloid is exactly what later creates a possible second branch.

Corollary 3.3 (Self-consistent frozen OT problem)**.** *A pair $(\pi_\star, A_\star)$ minimizes the joint problem in* (23) *if and only if $\pi_\star$ is GW-optimal,*

$$A_\star = 4C_{\pi_\star}, \tag{26}$$

and $\pi_\star$ solves ordinary Kantorovich transport for the cost $c_{A_\star}$. At self-consistency this cost can be evaluated only from the centered kernels:

$$c_{\pi_\star}(x,y) = -4a(x)b(y) - 16\int k_\alpha(x,x')k_\beta(y,y')\,\mathrm{d}\pi_\star(x',y'). \tag{27}$$

Proof. If $(\pi_\star, A_\star)$ is jointly optimal, minimizing first in A and using (22) gives $A_\star = 4C_{\pi_\star}$ and shows that $\pi_\star$ is GW-optimal. With $A_\star$ frozen, replacing $\pi_\star$ by a better linear-OT coupling would decrease the joint objective, so $\pi_\star$ solves that frozen problem. Conversely, if $\pi_\star$ is GW-optimal and $A_\star = 4C_{\pi_\star}$, then (22) makes the pair jointly optimal; its frozen optimality follows by the preceding replacement argument. Finally,

$$\langle C_{\pi_\star}u(x), v(y)\rangle = \int k_\alpha(x,x')k_\beta(y,y')\,\mathrm{d}\pi_\star(x',y'),$$

which gives (27). □

The corollary is the bridge to Monge theory: the nonlinear GW minimizer is a linear OT minimizer once its learned cost has been frozen. The remaining question is whether this particular cost forces graph support.

The freezing principle is considerably more general than the negative-type feature representation. Let Q be any continuous symmetric kernel on $(\mathcal{X} \times \mathcal{Y})^2$, and define

$$\mathcal{E}_Q(\pi) := \iint Q(z, z')\, \mathrm{d}\pi(z)\, \mathrm{d}\pi(z'), \qquad z = (x, y). \tag{28}$$

This includes a general GW distortion by taking $Q((x, y), (x', y')) = L(d_{\mathcal{X}}(x, x'), d_{\mathcal{Y}}(y, y'))$.

Proposition 3.4 (Universal frozen-cost and replacement principles)**.** *Let $\pi_\star$ minimize $\mathcal{E}_Q$ over $\Pi(\alpha, \beta)$, and set*

$$c_{\pi_\star}(z) := 2 \int Q(z, z')\, \mathrm{d}\pi_\star(z'). \tag{29}$$

Then $\pi_\star$ minimizes the linear OT problem with cost $c_{\pi_\star}$. This conclusion requires no convexity or concavity of $\mathcal{E}_Q$.

More precisely, if $\widetilde{\pi}$ is any minimizer of the frozen linear problem and $\eta = \widetilde{\pi} - \pi_\star$, then

$$\mathcal{E}_Q(\widetilde{\pi}) - \mathcal{E}_Q(\pi_\star) = \iint Q(z, z')\, \mathrm{d}\eta(z)\, \mathrm{d}\eta(z') \geq 0. \tag{30}$$

Consequently, $\widetilde{\pi}$ is GW-optimal if and only if the quadratic term in (30) *vanishes.*

Suppose, in addition, that

$$\iint Q(z, z')\, \mathrm{d}\eta(z)\, \mathrm{d}\eta(z') \leq 0 \quad \textit{for } \eta = \pi_1 - \pi_0, \quad \pi_0, \pi_1 \in \Pi(\alpha, \beta). \tag{31}$$

Equivalently, $\mathcal{E}_Q$ is concave on $\Pi(\alpha, \beta)$. Then every minimizer of the frozen linear problem for $c_{\pi_\star}$ is also a global minimizer of $\mathcal{E}_Q$. It is sufficient, but generally stronger, to require the same inequality for every finite signed η with zero $\mathcal{X}$- and $\mathcal{Y}$-marginals.

Proof. For $\pi \in \Pi(\alpha, \beta)$, put $\eta = \pi - \pi_\star$. Expanding along the feasible segment gives

$$\mathcal{E}_Q(\pi_\star + t\eta) = \mathcal{E}_Q(\pi_\star) + t \int c_{\pi_\star}\, \mathrm{d}\eta + t^2 \iint Q\, \mathrm{d}\eta\, \mathrm{d}\eta. \tag{32}$$

Minimality at $t = 0$ forces $\int c_{\pi_\star}\, \mathrm{d}\eta \geq 0$, proving frozen optimality. If $\widetilde{\pi}$ is another frozen minimizer, this linear term vanishes for $\eta = \widetilde{\pi} - \pi_\star$. Setting $t = 1$ proves the equality in (30); its inequality follows from global minimality of $\pi_\star$. Under (31), the expansion at $t = 1$ gives $\mathcal{E}_Q(\widetilde{\pi}) \leq \mathcal{E}_Q(\pi_\star)$. The reverse inequality follows from global minimality of $\pi_\star$, hence equality holds. □

The first assertion is a necessary first-order condition for every quadratic GW problem. The second is stronger: conditional concavity permits replacing an optimizer by a more structured optimizer of its frozen problem. This is the mechanism needed below to manufacture a Monge GW optimizer through a quotient, rather than merely diagnose whether a prescribed optimizer is a map. For power-distance GW, conditional concavity is exactly (15).

The same frozen problem admits a direct expression that does not require constructing the feature spaces. It is the first variation of the original quadratic GW objective after discarding terms whose integral is fixed by the marginals.

Proposition 3.5 (Direct first-variation cost)**.** *For $\pi \in \Pi(\alpha, \beta)$, define*

$$\ell_\pi(x, y) := -4 \int d_{p,\mathcal{X}}(x, x') d_{p,\mathcal{Y}}(y, y') \, \mathrm{d}\pi(x', y'). \tag{33}$$

If $A_\pi = 4C_\pi$, then there are continuous functions $\varphi_\pi : \mathcal{X} \to \mathbb{R}$, $\psi_\pi : \mathcal{Y} \to \mathbb{R}$, and a constant q_π such that

$$\ell_\pi(x, y) = c_{A_\pi}(x, y) + \varphi_\pi(x) + \psi_\pi(y) + q_\pi. \tag{34}$$

Consequently, ℓ_π and c_{A_π} have the same optimal couplings. Wherever both costs are differentiable in x, their twist maps differ by a vector independent of y, and hence have the same injectivity. In particular, every GW minimizer $\pi_\star$ solves linear OT for $\ell_{\pi_\star}$.

If $1 < p \leq 2$, differentiation under the integral gives

$$\nabla_x \ell_\pi(x, y) = -4p \int \|x - x'\|^{p-2} (x - x') \, \|y - y'\|^p \, \mathrm{d}\pi(x', y'), \tag{35}$$

where the vector integrand is set to zero at $x = x'$.

Proof. For a feasible signed direction η with zero marginals, the first variation is

$$D\mathcal{E}_p(\pi)[\eta] = 2 \iint \left(d_{p,\mathcal{X}}(x, x') - d_{p,\mathcal{Y}}(y, y')\right)^2 \mathrm{d}\pi(x', y') \, \mathrm{d}\eta(x, y).$$

Thus it is represented on the fixed-marginal affine set by the function

$$(x, y) \longmapsto 2 \int \left(d_{p,\mathcal{X}}(x, x') - d_{p,\mathcal{Y}}(y, y')\right)^2 \mathrm{d}\pi(x', y').$$

The two squared terms depend only on x or only on y; their integrals are fixed on $\Pi(\alpha, \beta)$. Removing them leaves (33). Alternatively, insert

$$d_{p,\mathcal{X}}(x, x') = a(x) + a(x') - 2k_\alpha(x, x')$$

and its $\mathcal{Y}$-counterpart into (33). The only terms depending jointly on (x, y) are

$$-4a(x)b(y) - 16 \int k_\alpha(x, x') k_\beta(y, y') \, \mathrm{d}\pi(x', y'),$$

which equal c_{A_π} by (27); all remaining terms are separable or constant. This proves (34). Such additions do not change a Kantorovich optimizer, and their x-gradients do not depend on y, so they do not change twist. The last assertion follows either from corollary 3.3 or by applying proposition 3.4 to the quadratic functional $\mathcal{E}_p$. Formula (35) follows by differentiating $\|x - x'\|^p$. □

4 The Brenier mechanism: twist of the learned cost

For general costs, the appropriate replacement for strict convexity of the quadratic cost is control of the twist fibers. One branch gives a Monge map; finitely many branches give a finite mixture of maps. Since the source is absolutely continuous, these conditions are needed only almost everywhere in the source variable and only on the dual contact set.

4.1 Finite twist and finite-map solutions

For a dual feasible pair (f, g), define its contact set

$$S_{f,g} := \{(x, y) : f(x) + g(y) = c(x, y)\}. \tag{36}$$

Every optimal plan is concentrated on this set when (f, g) is dual optimal. It is therefore unnecessary to control all fibers of the twist map; controlling the fibers that meet $S_{f,g}$ is enough.

Definition 4.1 (Finite twist on a contact set)**.** Let c be differentiable in its first variable and let $S \subset \mathcal{X} \times \mathcal{Y}$. The cost is *$m$-twisted on S at x* if, for every $q \in \mathbb{R}^n$,

$$\operatorname{card}\{y \in \operatorname{supp}\beta : (x, y) \in S,\ \nabla_x c(x, y) = q\} \le m. \tag{37}$$

It is *α-almost-everywhere m-twisted on S* if (37) holds for α-almost every x. The global m-twist condition is obtained by taking $S = \operatorname{supp}\alpha \times \operatorname{supp}\beta$; ordinary twist is the case $m = 1$.

This finite-fiber relaxation of twist is precisely the condition that yields a finite mixture of transport maps (Moameni and Pass, 2017).

Proposition 4.2 (Finite twist selects finitely many graphs)**.** *Assume $\alpha \ll \mathrm{Leb}^n$, the supports are compact, and c is continuous and C^1 in its first variable near the supports. Let (f, g) be a dual optimizer for (20). If c is α-almost-everywhere m-twisted on $S_{f,g}$, then every primal optimizer π is a k-map for some $k \le m$: there are measurable maps T_j and weights $\lambda_j \ge 0$, $\sum_{j=1}^{k} \lambda_j = 1$, such that*

$$\pi = \sum_{j=1}^{k} (\mathrm{Id}, T_j)_\sharp(\lambda_j \alpha). \tag{38}$$

At every differentiability point of f, each active branch satisfies

$$\nabla f(x) = \nabla_x c(x, T_j(x)) \qquad \text{whenever } \lambda_j(x) > 0. \tag{39}$$

Proof. The uniform bound on $\nabla_x c$ makes the c-concave potential f Lipschitz, hence differentiable α-almost everywhere. Disintegrate an optimal plan as $\pi = \int \delta_x \otimes \pi_x \, \mathrm{d}\alpha(x)$. At a differentiability point x, every $y \in \operatorname{supp}\pi_x$ lies in the contact set. Since $f(z) \le c(z, y) - g(y)$, with equality at $z = x$, differentiation gives $\nabla f(x) = \nabla_x c(x, y)$. Condition (37) therefore implies $\operatorname{card}(\operatorname{supp}\pi_x) \le m$. A finite-valued Borel multifunction admits a measurable enumeration; enumerating the atoms of π_x and taking their masses gives maps T_j and weights λ_j, after adding zero-weight branches where necessary. This proves (38) and (39). □

Corollary 4.3 (Twist selects a unique graph coupling)**.** *Under the assumptions of proposition 4.2, if c is α-almost-everywhere twisted on $S_{f,g}$, then (20) has a unique optimizer $(\mathrm{Id}, T)_\sharp\alpha$, characterized by*

$$\nabla f(x) = \nabla_x c(x, T(x)) \qquad \text{for } \alpha\text{-almost every } x. \tag{40}$$

Proof. The preceding proposition gives a graph optimizer. If two optimal graph couplings existed, their average would be optimal and the same argument would force its conditional measures to be Dirac masses. The two maps must therefore agree α-almost everywhere. □

Combining this linear theorem with the universal freezing principle gives the most general direct Brenier statement available for a quadratic GW energy.

Corollary 4.4 (Abstract frozen-cost Brenier theorem)**.** *Let Q be as in (28), assume $\alpha \ll \mathrm{Leb}^n$, and let $\pi_\star$ minimize $\mathcal{E}_Q$. Suppose that its frozen cost $c_{\pi_\star}$ is continuous and C^1 in its first variable near the compact supports. If this cost is m-twisted on a dual contact set for α-almost every source point, then $\pi_\star$ is a k-map for some $k \le m$. In particular, one-twist implies that $\pi_\star$ is a Monge coupling and is the unique optimizer of its frozen linear problem. No concavity of $\mathcal{E}_Q$ is needed.*

Proof. By proposition 3.4, $\pi_\star$ minimizes linear OT for $c_{\pi_\star}$. Apply proposition 4.2 and corollary 4.3. □

4.2 The finite-branch power-distance theorem

Freezing a GW optimizer now transfers the preceding linear result without loss. This is the appropriate Brenier statement when the twist map has more than one inverse branch.

Theorem 4.5 (Self-consistent finite-twist theorem)**.** *Let $0 < p \leq 2$, assume $\alpha \ll \mathrm{Leb}^n$, and let $\pi_\star$ minimize (2). Set $A_\star = 4C_{\pi_\star}$, let $(f_\star, g_\star)$ be a dual optimizer for the frozen cost $c_{A_\star}$, and suppose that this cost is C^1 in x near the supports. If $c_{A_\star}$ is α-almost-everywhere m-twisted on $S_{f_\star,g_\star}$, then $\pi_\star$ is a k-map for some $k \leq m$, with active branches characterized by*

$$\nabla f_\star(x) = \nabla_x c_{A_\star}(x, T_j(x)). \tag{41}$$

If $m = 1$, the frozen linear optimizer is unique and $\pi_\star$ is a Monge coupling.

Proof. By corollary 3.3, $\pi_\star$ solves linear OT for $c_{A_\star}$. Apply proposition 4.2; the final claim is corollary 4.3. □

Corollary 4.6 (RKHS-twist Brenier theorem for power-distance GW)**.** *Under the assumptions of theorem 4.5, suppose that $c_{A_\star}$ is α-almost-everywhere twisted. Then*

$$\pi_\star = (\mathrm{Id}, T_\star)_\sharp \alpha, \tag{42}$$

and it is the unique optimizer of the frozen problem. Moreover,

$$\nabla f_\star(x) = \nabla_x c_{A_\star}(x, T_\star(x)) \quad \textit{for } \alpha\textit{-almost every } x. \tag{43}$$

This is a genuine Brenier-type conclusion, but the relevant inverse can be multivalued: (41) selects the branches of $y \mapsto \nabla_x c_{A_\star}(x, y)$. Only under one-twist is there a single map, and that map need not be the gradient of an ordinary convex function.

For $1 < p \leq 2$, the twist map can be evaluated without constructing the Hilbert features. Define

$$G_{\pi,x}(y) := \int \|x - x'\|^{p-2} (x - x') \|y - y'\|^p \, \mathrm{d}\pi(x', y'). \tag{44}$$

By (35), the first-variation cost has twist map $-4pG_{\pi,x}$.

Corollary 4.7 (Distance-only finite-twist criterion)**.** *Assume $1 < p \leq 2$, $\alpha \ll \mathrm{Leb}^n$, and compact supports. Let $\pi_\star$ be GW-optimal. If there is $m < \infty$ such that, for α-almost every x and every $q \in \mathbb{R}^n$,*

$$\mathrm{card}\{y \in \mathrm{supp}\,\beta : G_{\pi_\star,x}(y) = q\} \leq m, \tag{45}$$

then $\pi_\star$ is a k-map for some $k \leq m$. For $m = 1$, it is a Monge coupling. It is enough that (45) hold after intersecting each fiber with a dual contact set of the frozen problem.

Proof. For $p > 1$, the derivative of $x \mapsto \|x - x'\|^p$ is continuous, including at $x = x'$, and uniformly bounded on compact sets. Thus $\ell_{\pi_\star}$ is continuously differentiable in x, and (35) identifies its gradient with $-4pG_{\pi_\star,x}$. Since proposition 3.5 makes $\pi_\star$ optimal for this frozen cost, proposition 4.2 applies. □

4.3 Quotient twist and transport inside the fibers

Finite twist is a theorem about a *given* optimal coupling. A second mechanism can prove existence of *some* Monge optimizer even when every twist fiber is infinite. It separates the variables seen by the cost from the directions along which the cost is indifferent. One first transports the visible quotient variables and then uses the invisible source fibers to reproduce the target conditional laws. The measurable lifting result below is the abstract mechanism developed by Dumont et al. (2025, Theorem 2.1).

Theorem 4.8 (Quotient-and-fiber lifting for linear OT)**.** *Let $\mathcal{X}, \mathcal{Y}, \mathcal{U}, \mathcal{V}$ be standard Borel spaces, let $\phi : \mathcal{X} \to \mathcal{U}$ and $\psi : \mathcal{Y} \to \mathcal{V}$ be measurable, and suppose that*

$$c(x, y) = \widetilde{c}\big(\phi(x), \psi(y)\big). \tag{46}$$

Set $\bar{\alpha} = \phi_\sharp \alpha$ and $\bar{\beta} = \psi_\sharp \beta$. Assume that quotient OT for $\widetilde{c}$ between $\bar{\alpha}$ and $\bar{\beta}$ has an optimal map $t : \mathcal{U} \to \mathcal{V}$, and let

$$\alpha = \int \alpha_u \, \mathrm{d}\bar{\alpha}(u), \qquad \beta = \int \beta_v \, \mathrm{d}\bar{\beta}(v) \tag{47}$$

be disintegrations along ϕ and ψ. If α_u is nonatomic for $\bar{\alpha}$-almost every u, or more generally if there is a jointly measurable family of maps satisfying

$$(T_u)_\sharp \alpha_u = \beta_{t(u)} \quad \text{for } \bar{\alpha}\text{-almost every } u, \tag{48}$$

then there exists a measurable map $T : \mathcal{X} \to \mathcal{Y}$ such that

$$T_\sharp \alpha = \beta, \qquad \psi\big(T(x)\big) = t\big(\phi(x)\big) \quad \text{for } \alpha\text{-almost every } x, \tag{49}$$

and $(\mathrm{Id}, T)_\sharp \alpha$ is optimal for the original cost c.

Proof. When the fiber maps are not given directly, the measurable randomization lemma for atomless probability kernels supplies them. Concretely, one may measurably identify every atomless α_u with Lebesgue measure on $[0, 1]$, and use a measurable quantile parametrization of the kernel $v \mapsto \beta_v$; the joint measurability is the nontrivial part of the lifting theorem. Set $T(x) = T_{\phi(x)}(x)$. Then

$$\begin{aligned} T_\sharp \alpha &= \int (T_u)_\sharp \alpha_u \, \mathrm{d}\bar{\alpha}(u) = \int \beta_{t(u)} \, \mathrm{d}\bar{\alpha}(u) \\ &= \int \beta_v \, \mathrm{d}\bar{\beta}(v) = \beta, \end{aligned}$$

because $t_\sharp \bar{\alpha} = \bar{\beta}$. This also proves the projection identity in (49).

For any $\pi \in \Pi(\alpha, \beta)$, the pushforward $(\phi, \psi)_\sharp \pi$ belongs to $\Pi(\bar{\alpha}, \bar{\beta})$, and

$$\int c \, \mathrm{d}\pi = \int \widetilde{c} \, \mathrm{d}\big((\phi, \psi)_\sharp \pi\big).$$

The graph coupling constructed above projects to $(\mathrm{Id}, t)_\sharp \bar{\alpha}$, so it attains the quotient optimum and is therefore optimal for c. □

Atomlessness is therefore a clean sufficient condition, not a logically necessary one. For example, no atomlessness is needed when $\beta_{t(u)}$ is a Dirac mass. Conversely, if both conditional laws have incompatible atoms, quotient optimality need not lift to a map. This fiber compatibility is exactly what distinguishes a quotient argument from a twist argument.

The replacement principle in proposition 3.4 now turns this linear lifting theorem into an abstract GW Brenier theorem.

Theorem 4.9 (Quotient-and-fiber Brenier principle for GW)**.** *Let Q satisfy the conditional concavity condition (31), let $\pi_\star$ minimize (28), and let $c_{\pi_\star}$ be its frozen cost (29). Suppose that this cost admits a factorization (46) for which the hypotheses of theorem 4.8 hold. Then $\mathcal{E}_Q$ has a Monge minimizer $(\mathrm{Id}, T)_\sharp \alpha$.*

If $\mathcal{U}, \mathcal{V}$ are compact subsets of $\mathbb{R}^r$, the quotient cost is continuous, continuously differentiable in its first variable near the supports, and twisted there, and $\bar{\alpha} \ll \mathrm{Leb}^r$, then the required quotient map exists and is unique.

Proof. By proposition 3.4, $\pi_\star$ minimizes linear OT for $c_{\pi_\star}$. The quotient-and-fiber theorem constructs a graph coupling $\widehat{\pi} = (\mathrm{Id}, T)_\sharp \alpha$ that minimizes the same frozen problem. Conditional concavity and the replacement part of proposition 3.4 imply that $\widehat{\pi}$ is also a global minimizer of $\mathcal{E}_Q$. The final assertion is the usual twisted-cost Monge theorem applied on the quotient spaces. □

The concavity hypothesis is essential for this *replacement* argument. Without it, the original optimizer $\pi_\star$ still solves its frozen linear problem, but another optimizer of that linear problem need not remain GW-optimal. Thus one-twist of the full frozen cost proves that $\pi_\star$ itself is a map without concavity, whereas quotient lifting proves existence of a possibly different map by using concavity. For a particular lifted coupling $\widehat{\pi}$, global conditional concavity can be weakened to the exact directional condition

$$\iint Q(z, z')\, \mathrm{d}(\widehat{\pi} - \pi_\star)(z)\, \mathrm{d}(\widehat{\pi} - \pi_\star)(z') = 0, \tag{50}$$

by (30). Condition (31) is valuable because it enforces this identity automatically for every frozen lift. Nothing in theorem 4.9 requires a linear or finite-rank factorization: any finite-dimensional sufficient statistic (ϕ, ψ) for the frozen cost is admissible. Finite rank is treated next because it supplies such statistics canonically for the learned RKHS cost.

4.4 Finite-rank quotient lifting and feature slices

Condition (45) is exact but still depends on the unknown optimizer. Finite rank gives two complementary routes. Its quotient variables can produce a Monge optimizer by the preceding lifting theorem, while their affine slices control the number of branches of a prescribed optimizer.

Suppose first that the learned operator has finite rank R, with singular value decomposition

$$A = \sum_{j=1}^{R} \sigma_j e_j \otimes f_j, \qquad \varphi_j(x) := \langle u(x), f_j \rangle, \qquad \psi_j(y) := \langle v(y), e_j \rangle. \tag{51}$$

Define the visible source and target statistics

$$\Phi_A(x) := \big(a(x), \varphi_1(x), \dots, \varphi_R(x)\big), \qquad \Psi_A(y) := \big(b(y), \psi_1(y), \dots, \psi_R(y)\big), \tag{52}$$

$$D_A := \mathrm{diag}(1, \sigma_1, \dots, \sigma_R). \tag{53}$$

Then the learned cost factors through an $(R+1)$-dimensional bilinear problem:

$$c_A(x, y) = -4 \langle D_A \Phi_A(x), \Psi_A(y) \rangle . \tag{54}$$

Theorem 4.10 (Finite-rank quotient Brenier theorem)**.** *Let $0 < p \le 2$, let $\pi_\star$ be GW-optimal, and suppose that $A_\star = 4C_{\pi_\star}$ has finite rank R. Assume that*

$$(\Phi_{A_\star})_\sharp \alpha \ll \mathrm{Leb}^{R+1} \tag{55}$$

and that the conditional measures in a disintegration of α along $\Phi_{A_\star}$ are nonatomic almost everywhere. Then there exists a GW-optimal map $T_\star : \mathcal{X} \to \mathcal{Y}$. It may be chosen so that

$$\Psi_{A_\star}\big(T_\star(x)\big) = t_\star\big(\Phi_{A_\star}(x)\big), \tag{56}$$

where $t_\star$ is the unique quotient OT map for the bilinear cost in (54). Moreover, the lifted optimizer has the same cross-covariance operator as $\pi_\star$.

Proof. All singular values in (53) are positive, so $(D_{A_\star})_\sharp(\Phi_{A_\star})_\sharp\alpha$ has a density. Moreover,

$$-4\langle D_{A_\star}s, z\rangle = 2\|D_{A_\star}s - z\|^2 - 2\|D_{A_\star}s\|^2 - 2\|z\|^2.$$

The last two terms are fixed by the quotient marginals, so the classical Brenier theorem gives a unique quotient map $t_\star$; equivalently, $t_\star(s) = \nabla h(D_{A_\star}s)$ for a convex function h. The factorization (54) and the concavity established in (15) allow theorem 4.9 to give a GW-optimal lift.

It remains to verify the last assertion. Put $\eta = (\mathrm{Id}, T_\star)_\sharp\alpha - \pi_\star$. Extend (13) linearly to signed measures, so that $C_\eta = C_{(\mathrm{Id},T_\star)_\sharp\alpha} - C_{\pi_\star}$. The two couplings tie for the frozen linear objective and for the GW objective. By (15), the quadratic coefficient of $t \mapsto \mathcal{E}_p(\pi_\star + t\eta)$ is $-8\left\|C_\eta\right\|_{\mathrm{HS}}^2$. The segment expansion (32) consequently forces $C_\eta = 0$, proving $C_{(\mathrm{Id},T_\star)_\sharp\alpha} = C_{\pi_\star}$. □

The two measure-theoretic hypotheses have a useful differential sufficient condition.

Corollary 4.11 (A coarea criterion)**.** *In addition to the assumptions of theorem 4.10, suppose that $\mathcal{X} \subset \mathbb{R}^n$, $\alpha \ll \mathrm{Leb}^n$, $\Phi_{A_\star}$ is C^1 near* $\operatorname{supp}\alpha$, *and*

$$n > R+1, \qquad \operatorname{rank} D\Phi_{A_\star}(x) = R+1 \quad \text{for } \alpha\text{-almost every } x. \tag{57}$$

Then a GW-optimal Monge map exists.

Proof. The coarea formula implies that $(\Phi_{A_\star})_\sharp\alpha$ has a density on $\mathbb{R}^{R+1}$. It also represents the conditional source measures by densities with respect to $\mathcal{H}^{n-R-1}$ on the regular fibers of $\Phi_{A_\star}$. Since $n - R - 1 \geq 1$, these conditional measures are nonatomic. Apply theorem 4.10. □

The coarea criterion is not a finite-twist condition: its regular fibers have positive dimension. Those fibers are precisely the reservoir of randomness used to match the target conditional laws deterministically.

Whenever a and the φ_j are differentiable, define

$$Z_A(y) := \Psi_A(y) \in \mathbb{R}^{R+1}, \tag{58}$$

$$B_{A,x}(s_0,\ldots,s_R) := s_0\nabla a(x) + \sum_{j=1}^{R} s_j\sigma_j\nabla\varphi_j(x) \in \mathbb{R}^n. \tag{59}$$

Proposition 4.12 (Finite-rank feature-slice criterion)**.** *Under the preceding assumptions, the learned twist map factors as*

$$\nabla_x c_A(x,y) = -4B_{A,x}Z_A(y). \tag{60}$$

Consequently, c_A is m-twisted at x whenever every affine fiber of $B_{A,x}$ intersects $Z_A(\mathcal{Y})$ in at most m points after pulling the intersection back through Z_A. If, in addition, Z_A has fibers of cardinality at most s, the complex projective closure of $Z_A(\mathcal{Y})$ is an algebraic variety of degree D, and these linear sections are zero dimensional, then one may take $m \leq sD$.

Proof. The singular value decomposition gives $\langle Au(x), v(y)\rangle = \sum_j \sigma_j\varphi_j(x)\psi_j(y)$. Differentiating (21) in x proves (60). The first conclusion is then just a fiber identity. The algebraic conclusion is the definition of degree: a zero-dimensional linear section of a projective variety contains at most D complex points when multiplicities are counted. Each image point has at most s preimages under Z_A, which gives sD. □

For $p = 2$ and full row rank, the target feature set is, after an invertible linear change of coordinates, the paraboloid

$$Z_A(y) = \left(\left\|y - m_\beta\right\|^2, y - m_\beta\right), \tag{61}$$

whose projective closure has degree two. The finite-rank criterion therefore predicts at most two branches; lemma 5.1 below verifies the zero-dimensionality condition and locates exactly where the second branch can occur. For $0 < p < 2$, the optimal Hilbert–Schmidt operator need not have finite rank, and even a finite-rank truncation does not preserve exact optimality. Thus this argument identifies a genuine rigidity route, but not an automatic finite-map theorem.

For quadratic dissimilarities, the coarea criterion recovers the low-rank Monge construction and sharpens it when the ambient source has additional dimensions.

Corollary 4.13 (Quadratic finite-rank quotient regimes). *Let $p = 2$, let $n \geq d$, assume $\alpha \ll \mathrm{Leb}^n$, and let $\pi_\star$ be GW-optimal with*

$$h := \operatorname{rank} S_{\pi_\star}. \tag{62}$$

If $n > h + 1$, then there exists a GW-optimal Monge map with the same cross-covariance as $\pi_\star$. Consequently, this conclusion holds whenever either

$$h \leq d - 2, \qquad \text{or} \qquad h = d - 1 \ \text{ and } \ n > d. \tag{63}$$

Proof. Up to orthogonal coordinates, the source quotient statistics are

$$\Phi_{A_\star}(x) = \left(\|x - m_\alpha\|^2, \langle x - m_\alpha, f_1\rangle, \ldots, \langle x - m_\alpha, f_h\rangle\right), \tag{64}$$

where $f_1, \ldots, f_h$ are orthonormal. Its differential has rank $h+1$ unless $x - m_\alpha \in \operatorname{span}\{f_1, \ldots, f_h\}$. This exceptional affine subspace is α-null. If $n > h+1$, corollary 4.11 applies. Finally, $h \leq d-2$ and $n \geq d$ imply $n > h+1$, while $h = d-1$ and $n > d$ give the same strict inequality. □

For $h \leq d-2$, this is the quotient mechanism behind the map theorem of Dumont et al. (2025). When $h = d-1$, equal dimensions leave zero-dimensional source and target fibers whose atomic weights need not be compatible; the general theorem then gives only two branches. The strict dimension gap turns the source fibers into positive-dimensional spheres, so they become nonatomic and a single-map lift exists. Full row rank $h = d$ is treated more efficiently by the direct twist argument in theorem 5.2, including the borderline case $n = d+1$.

4.5 Differential, generic, and definable finite-twist criteria

Proposition 4.14 (Mixed-Hessian immersion gives finite twist). *Let $\mathcal{X} \subset \mathbb{R}^n$ be compact and let $\mathcal{Y}$ be a compact d-dimensional C^1 manifold, possibly with boundary. Suppose $F(x, y) := \nabla_x c(x, y)$ is continuous and continuously differentiable in y, and*

$$\operatorname{rank} D_y F(x, y) = d \qquad \text{for every } (x, y) \in \mathcal{X} \times \mathcal{Y}. \tag{65}$$

Then there is an integer $m < \infty$ such that every fiber of $y \mapsto F(x, y)$ has at most m points, uniformly in x. Hence c is globally m-twisted.

Proof. At each (x_0, y_0), choose d output coordinates for which the corresponding $d \times d$ minor of $D_y F(x_0, y_0)$ is invertible. The inverse-function theorem with parameters gives product neighborhoods $U \times V$ of (x_0, y_0) on which, for every fixed $x \in U$, the restriction $F_x|_V$ is injective. Compactness provides a finite cover by such product neighborhoods. A fiber meets each member of this cover in at most one point, so its cardinality is bounded by the number of members in the cover. □

For the power-distance cost, the mixed Hessian has a particularly transparent form. With vectors understood as columns, set

$$M_{\pi,p}(x, y) := \int \|x - x'\|^{p-2} \|y - y'\|^{p-2} (x - x')(y - y')^\top \, \mathrm{d}\pi(x', y') \in \mathbb{R}^{n \times d}, \tag{66}$$

using the continuous zero extension of $z \mapsto \|z\|^{p-2} z$ at $z = 0$.

Corollary 4.15 (A cross-Hessian criterion for finite-branch GW). *Let $1 < p \le 2$, let $\alpha \ll \mathrm{Leb}^n$, and suppose the compact target support lies in a compact d-dimensional C^1 manifold. If a GW optimizer $\pi_\star$ satisfies*

$$\operatorname{rank} M_{\pi_\star,p}(x,y) = d \qquad \text{on } \operatorname{supp}\alpha \times \operatorname{supp}\beta, \tag{67}$$

then $\pi_\star$ is a k-map for some finite k.

Proof. Differentiating (44) in y gives

$$D_y G_{\pi,x}(y) = p M_{\pi,p}(x,y), \qquad D_y \nabla_x \ell_\pi(x,y) = -4p^2 M_{\pi,p}(x,y). \tag{68}$$

The integrand is continuous for $p > 1$, so differentiation under the integral is valid on compact sets. Apply proposition 4.14 and corollary 4.7. □

Condition (67) is strong: it rules out folds of the projected feature geometry. A weaker, generic mechanism counts how many sheets a smooth map can have. For fixed x, write $F_x(y) = \nabla_x c(x,y)$, and for pairwise distinct $(y_1,\dots,y_r)$ define the coincidence map

$$\Delta_r F_x(y_1,\dots,y_r) := \big(F_x(y_2) - F_x(y_1), \dots, F_x(y_r) - F_x(y_1)\big) \in (\mathbb{R}^n)^{r-1}. \tag{69}$$

Proposition 4.16 (General-position branch count). *Assume the hypotheses of proposition 4.2, let $n > d$, let $\mathcal{Y}$ be a compact smooth d-manifold, and suppose $F_x(y) = \nabla_x c(x,y)$ is smooth in y. Set*

$$m_{n,d} := \left\lfloor \frac{n}{n-d} \right\rfloor. \tag{70}$$

If, for α-almost every x, the map $\Delta_{m_{n,d}+1} F_x$ is transverse to the origin on the set of pairwise distinct tuples, then every fiber of F_x has at most $m_{n,d}$ points. Consequently, a frozen optimal plan for c is an $m_{n,d}$-map. In particular, general position predicts one branch when $n > 2d$, at most two when $n = 2d$, and at most three when $(n,d) = (3,2)$.

Proof. The domain of $\Delta_r F_x$ has dimension rd, whereas its target has dimension $(r-1)n$. For $r = m_{n,d} + 1$,

$$rd - (r-1)n = d - m_{n,d}(n-d) < 0.$$

A map from a lower-dimensional manifold cannot be transverse to the origin at a preimage of the origin. Hence no r distinct points have the same image, which is exactly the asserted fiber bound. The finite-map conclusion follows from proposition 4.2. □

The restriction $n > d$ is structural for this argument. If $n < d$, a regular fiber of a smooth map $\mathcal{Y}^d \to \mathbb{R}^n$ has dimension $d - n$, so finite twist is nongeneric. If $n = d$, regular fibers are discrete but their number is a topological or algebraic degree not controlled by dimension alone.

Multijet transversality makes the hypothesis of proposition 4.16 generic for unconstrained smooth maps (Golubitsky and Guillemin, 1973). It is not automatically generic inside the much smaller family of self-consistent GW costs. Proving transversality with respect to perturbations of (α,β), while retaining the fixed-point relation $A_\star = 4C_{\pi_\star}$, is a concrete route to a generic finite-map theorem. For $1 < p < 2$, smoothness of $G_{\pi_\star,x}$ itself requires regularity of the vector-valued measures obtained by weighting $\pi_\star$; smoothness of β alone does not automatically provide it.

Analyticity offers a different rigidity mechanism, but only after positive-dimensional fibers have been excluded.

Proposition 4.17 (Definable analytic rigidity). *Let $\mathcal{X}$, $\mathcal{Y}$, and $F : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}^n$ be definable in a fixed o-minimal structure, with $\mathcal{Y}$ compact. Define*

$$E := \{x \in \mathcal{X} : \text{some fiber } \{y \in \mathcal{Y} : F(x, y) = q\} \text{ is infinite}\}. \tag{71}$$

Then E is definable and there exists $m < \infty$ such that

$$\operatorname{card}\{y \in \mathcal{Y} : F(x, y) = q\} \le m \qquad (x \notin E,\ q \in \mathbb{R}^n). \tag{72}$$

In particular, if $F(x, y) = \nabla_x c(x, y)$ and $\alpha(E) = 0$, then c is α-almost-everywhere m-twisted. This applies to a real-analytic cost on neighborhoods of compact semianalytic supports, provided no gradient fiber has positive dimension outside an α-null set.

Proof. In an o-minimal structure, an infinite definable fiber has positive dimension, and the locus where the fiber dimension is positive is definable. On the complement of this locus all fibers are finite. The uniform finiteness theorem for definable families then supplies a common cardinality bound (van den Dries, 1998). Restricted real-analytic functions on compact semianalytic sets are globally subanalytic, hence definable in $\mathbb{R}_{\mathrm{an}}$. □

There are two important qualifications. First, analyticity alone neither excludes infinite fibers nor fixes their number. On $[-1, 1]^2$, the analytic costs $c_N(x, y) = xT_N(y)$ built from Chebyshev polynomials satisfy $\partial_x c_N = T_N(y)$, whose typical fibers have N points; a cost independent of y has infinite fibers. Second, when $0 < p < 2$, the kernel $\|y - y'\|^p$ is not analytic on the diagonal, and integration against an arbitrary optimal coupling does not preserve definability. Thus proposition 4.17 is a conditional rigidity tool, not an automatic consequence of the Schoenberg representation.

In algebraic models the branch number can sometimes be made explicit. If the components of $F_x(y)$ are polynomials of degree at most r and every complexified fiber is zero dimensional, Bézout's theorem gives the crude uniform bound r^d. For $p = 2$, the frozen twist map is quadratic in y, so this argument gives 2^d; the rank-one structure analyzed next improves it sharply to two.

5 The finite-dimensional case $p = 2$

At $p = 2$, the abstract feature objects become ordinary centered moments. This permits an exact analysis of the obstruction.

5.1 The learned finite-dimensional cost

Write

$$\widetilde{x} = x - m_\alpha, \qquad \widetilde{y} = y - m_\beta. \tag{73}$$

Then

$$u(x) = \widetilde{x}, \quad v(y) = \widetilde{y}, \quad a(x) = \|\widetilde{x}\|^2, \quad b(y) = \|\widetilde{y}\|^2, \quad C_\pi = S_\pi.$$

The general cross-Hessian (66) reduces to

$$M_{\pi,2}(x, y) = \widetilde{x}\widetilde{y}^\top + S_\pi^\top. \tag{74}$$

The operator A is a matrix in $\mathbb{R}^{d\times n}$, the RKHS penalty is the Frobenius norm, and

$$c_A(x, y) = -4\,\|\widetilde{x}\|^2\,\|\widetilde{y}\|^2 - 4\,\langle A\widetilde{x}, \widetilde{y}\rangle\,, \qquad A_\star = 4S_{\pi_\star}. \tag{75}$$

The first term is the crucial difference from inner-product GW. Without it, full row rank of A immediately gives twist. With it, the gradient is

$$\nabla_x c_A(x, y) = -8\widetilde{x}\,\|\widetilde{y}\|^2 - 4A^\top\widetilde{y}. \tag{76}$$

5.2 One branch off the learned row space, two branches on it

Lemma 5.1 (Rank geometry of the squared-distance learned cost). *Assume $A \in \mathbb{R}^{d\times n}$ has full row rank. For every fixed x, each fiber of $y \mapsto \nabla_x c_A(x,y)$ contains at most two points. If*

$$\widetilde{x} \notin \operatorname{Ran}(A^\top), \tag{77}$$

then every fiber contains at most one point, so the cost is twisted at x.

Proof. Suppose two targets y_0, y have the same gradient. With $z_0 = y_0 - m_\beta$, $z = y - m_\beta$, and $r = \|z\|^2 - \|z_0\|^2$, (76) gives

$$A^\top(z - z_0) = -2r\widetilde{x}. \tag{78}$$

If $\widetilde{x} \notin \operatorname{Ran}(A^\top)$, the two sides can agree only when $r = 0$. Since full row rank makes $A^\top$ injective, this forces $z = z_0$.

It remains to count fibers when $\widetilde{x} \in \operatorname{Ran}(A^\top)$. There is a unique $q \in \mathbb{R}^d$ such that $A^\top q = -2\widetilde{x}$, and (78) gives $z = z_0 + rq$. Substitution into the definition of r yields

$$r = 2r\langle z_0, q\rangle + r^2\|q\|^2.$$

Besides $r = 0$, this scalar equation has at most one solution. Hence the gradient fiber has cardinality at most two. □

Thus the full-row-rank frozen cost is globally two-twisted, and proposition 4.2 turns any optimizer of that linear problem into a two-map. The following dimension gap strengthens two-twist to one-twist at α-almost every source point.

The exceptional set in (77) is a proper affine subspace as soon as the source dimension is strictly larger than the target dimension. This yields a concrete Monge theorem.

Theorem 5.2 (Dimension-gap Brenier theorem for $p = 2$). *Let $n > d$, let $\alpha \ll \mathrm{Leb}^n$, and assume compact supports. Let $\pi_\star$ be a minimizer of the squared-distance problem (2) with $p = 2$. If*

$$\operatorname{rank} S_{\pi_\star} = d, \tag{79}$$

then $\pi_\star$ is induced by a map: $\pi_\star = (\mathrm{Id}, T_\star)_\sharp\alpha$. More precisely, $\pi_\star$ is the unique optimizer of linear OT for $c_{A_\star}$, where $A_\star = 4S_{\pi_\star}$. If $f_\star$ is the corresponding source potential, then

$$\nabla f_\star(x) = -8(x - m_\alpha)\left\|T_\star(x) - m_\beta\right\|^2 - 16S_{\pi_\star}^\top(T_\star(x) - m_\beta) \tag{80}$$

for α-almost every x.

Proof. By corollary 3.3, $\pi_\star$ solves linear OT for the cost (75) with $A_\star = 4S_{\pi_\star}$. Assumption (79) makes $A_\star$ full row rank. The subspace $\operatorname{Ran}(A_\star^\top)$ has dimension $d < n$, so

$$\alpha\big(m_\alpha + \operatorname{Ran}(A_\star^\top)\big) = 0.$$

By lemma 5.1, the frozen cost is therefore α-almost-everywhere twisted. Apply corollary 4.3. Substituting $A_\star = 4S_{\pi_\star}$ into (76) gives (80). □

This conclusion is stronger than the generic count in proposition 4.16. When $d < n \le 2d$, that count allows several branches, whereas the special rank-one quadratic structure of (76) confines all branching to the d-dimensional affine space $m_\alpha + \operatorname{Ran}(A_\star^\top)$, which is α-null.

The dimension gap is not the only way to remove the second branch. If the target feature radius is constant, the nonlinear radial term in (75) becomes marginal-only and the learned cost is effectively bilinear.

Corollary 5.3 (Centered spherical targets)**.** *Let $n \geq d$, assume $\alpha \ll \mathrm{Leb}^n$, and let $\pi_\star$ be a minimizer for $p = 2$ with* $\operatorname{rank} S_{\pi_\star} = d$*. If, for some $R \geq 0$,*

$$\left\|y - m_\beta\right\| = R \qquad \textit{for every } y \in \operatorname{supp} \beta, \tag{81}$$

then $\pi_\star = (\mathrm{Id}, T_\star)_\sharp \alpha$, including when $n = d$.

Proof. On $\operatorname{supp} \beta$, (76) becomes

$$\nabla_x c_{A_\star}(x, y) = -8R^2(x - m_\alpha) - 4A_\star^\top(y - m_\beta).$$

The first term is independent of y, and full row rank makes $A_\star^\top$ injective. The frozen cost is therefore twisted at every x, so corollary 4.6 applies. □

Remark 5.4 (What the theorem does and does not assume)*.* The rank conditions in theorem 5.2 and corollary 5.3 are imposed on an optimal cross-covariance, hence they are not a priori conditions on the marginals alone. Their conclusions nevertheless concern the original GW optimizer, not a Gaussian or affine restriction. Under the strict dimension gap of theorem 5.2, if every optimal cross-covariance has full row rank, then every GW optimizer is graph-supported; the same statement holds under the spherical-target hypothesis of corollary 5.3. Full row rank alone is not claimed to suffice in equal dimensions for a general target.

Combining this observation with the known fiber construction gives the following rank map.

Corollary 5.5 (Known and new single-map regimes at $p = 2$)**.** *Under the assumptions of theorem 1.4, the following regimes are available.*

(i) If an optimal cross-covariance has rank at most $d - 2$, the fiber theorem of Dumont et al. (2025), recovered abstractly by corollary 4.13, provides an optimal map.

(ii) If $n > d$ and an optimal cross-covariance has rank $d - 1$, corollary 4.13 also provides an optimal map because the source quotient fibers are nonatomic.

(iii) If $n > d$ and an optimal cross-covariance has rank d, theorem 5.2 shows that this optimizer is a map.

(iv) If the centered target radius is constant and an optimal cross-covariance has rank d, corollary 5.3 gives a map even when $n = d$.

(v) When $n = d$ and the rank is $d - 1$, the explicit fiber analysis of Dumont et al. (2025) gives an optimal two-map, but no general one-map conclusion is known.

For $n = d$, full row rank does not yield a single branch because $\operatorname{Ran}(A^\top) = \mathbb{R}^n$*.*

5.3 Why equal dimensions retain two branches

When $n = d$ and A is invertible, the second part of lemma 5.1 applies at every source point. The scalar quadratic for r can have two roots, so the frozen cost is generally only two-twisted. Complementary slackness then restricts each conditional plan to at most two targets but does not force it to be a Dirac mass. This is the local mechanism behind theorem 1.4.

The distinction with inner-product GW is now exact. Its learned cost is bilinear, so the gradient equation is linear in y, and full rank gives a single inverse. Squared-distance GW adds the radial statistic $\left\|y - m_\beta\right\|^2$; solving simultaneously for the linear feature and that radial statistic produces a quadratic scalar equation. An RKHS norm penalty controls the learned cost but cannot remove this algebraic second branch.

6 What remains for $0 < p < 2$

The learned-cost theorem is exact for every $0 < p \leq 2$, but the sharp two-branch algebra used at $p = 2$ does not survive unchanged. The preceding finite-twist and quotient-lifting frameworks narrow the missing step without pretending that regularity alone solves it.

The feature lift is not a free Brenier theorem. For $0 < p < 2$, the Schoenberg feature space is generally infinite dimensional. Even when α has a smooth density in $\mathbb{R}^n$, its feature pushforward $\Phi_\sharp\alpha$ lies on a nonlinear finite-dimensional subset of $\mathcal{H}_{p,n}$. It is not absolutely continuous with respect to an infinite-dimensional analogue of Lebesgue measure. Applying a Hilbert-space version of Brenier's theorem directly to the feature measures is therefore unjustified. The finite-rank quotient theorem avoids this mistake by retaining only $R + 1$ visible statistics and by explicitly transporting within the discarded fibers. It does not apply automatically when the optimal operator has infinite rank.

There are two targets: finite twist or a liftable quotient. For a given optimizer, (44) is the explicit twist object to study. Uniformly bounding its contact fibers yields a finite-map optimizer by corollary 4.7; injectivity is only the special case that yields a Monge map. Alternatively, a finite-dimensional factorization of the frozen cost can produce a Monge optimizer through theorem 4.9, even when the full twist fibers are positive dimensional. The cross-Hessian condition (67) proves such finiteness by local immersion and compactness. More flexibly, proposition 4.16 gives the candidate generic count $m_{n,d}$ when $n > d$. What remains is to derive either finite twist or a finite-dimensional liftable quotient from assumptions on the marginals while respecting self-consistency of $\pi_\star$.

Equal dimensions require rigidity beyond dimension counting. When $n = d$, multijet dimension counting gives no finite universal branch number. Polynomial degree controls the $p = 2$ model, but for $0 < p < 2$ the frozen cost is generally neither polynomial nor analytic on the diagonal. The definable criterion of proposition 4.17 becomes effective only after one proves that positive-dimensional fibers are absent. Analyticity by itself cannot provide a dimension-only value of m, as the Chebyshev example shows.

The nonsmooth range needs a subdifferential version. For $1 < p < 2$, the mixed derivative in (68) remains continuous and the finite twist program is directly available. At $p \leq 1$, the power distance is nondifferentiable or singular at coincidences. One must replace $\nabla_x c$ by a suitable subdifferential fiber and control the resulting set-valued contact relation; this note does not yet prove that extension.

The endpoint $p > 2$ requires another mechanism. Euclidean p-powers cease to be conditionally negative definite in general when $p > 2$ (Schoenberg, 1938). The Hilbert feature representation, the Hilbert–Schmidt cross-covariance, and the quadratic RKHS regularizer used here then disappear. A Monge theorem in that range would need a different factorization or a direct analysis of the frozen GW cost.

Conclusion

The strongest rigorous narrative is therefore not that the RKHS reduction automatically proves a universal Brenier theorem. It proves something more precise:

1. for $0 < p \leq 2$, power-distance GW is exactly regularized learning of a cross-space cost in a tensor-product RKHS;
2. every GW optimizer solves ordinary OT for its self-consistent learned cost;
3. for every quadratic GW kernel, frozen optimality is a universal first-order condition; conditional concavity upgrades it to a replacement principle for all frozen optimizers;

4. an m-twisted contact relation forces a prescribed optimizer to be a measurable mixture of at most m maps, while one-twist gives a unique frozen graph coupling;
5. a twisted quotient problem together with nonatomic source fibers yields a GW-optimal graph coupling even when the full cost has infinite twist fibers;
6. for a rank-R learned operator, this quotient has only $R + 1$ coordinates, and the coarea criterion $n > R + 1$ makes its liftability explicit;
7. full column rank of the power cross-Hessian gives some finite branch number, and fiberwise general position predicts the explicit count $m_{n,d} = \lfloor n/(n-d) \rfloor$ when $n > d$;
8. for $p = 2$, quotient lifting recovers the low-rank map theorem and gives a map at rank $d - 1$ when $n > d$, while the special rank-one quadratic structure improves the generic count to two and, with a strict dimension gap and full row rank, to one almost everywhere;
9. a centered spherical target removes the radial branch and gives a map under full row rank even in equal dimensions; and
10. otherwise, in equal dimensions the radial feature can leave two branches, consistently with the known two-map theorem.

The remaining research problem for $1 < p < 2$ is therefore concrete: prove nondegeneracy, transversality, finite definable fibers, or a finite-rank liftable quotient for the self-consistent field $G_{\pi_\star,x}$. A one-map theorem through twist requires injectivity; quotient lifting instead requires sufficient conditional mass inside the invisible fibers.

References

Yann Brenier. Polar factorization and monotone rearrangement of vector-valued functions. *Communications on Pure and Applied Mathematics*, 44(4):375–417, 1991. doi: 10.1002/cpa.3160440402.

Théo Dumont, Théo Lacombe, and François-Xavier Vialard. On the existence of monge maps for the gromov–wasserstein problem. *Foundations of Computational Mathematics*, 25(2):463–510, 2025. doi: 10.1007/s10208-024-09643-0.

Martin Golubitsky and Victor Guillemin. *Stable Mappings and Their Singularities*, volume 14 of *Graduate Texts in Mathematics*. Springer, New York, 1973. doi: 10.1007/978-1-4615-7904-5.

Hiroshi Konno. Maximization of a convex quadratic function under linear constraints. *Mathematical Programming*, 11(1):117–127, 1976. doi: 10.1007/BF01580380.

Facundo Mémoli and Tom Needham. Comparison results for gromov–wasserstein and gromov–monge distances. *ESAIM: Control, Optimisation and Calculus of Variations*, 30:78, 2024. doi: 10.1051/cocv/2024063.

Abbas Moameni and Brendan Pass. Solutions to multi-marginal optimal transport problems concentrated on several graphs. *ESAIM: Control, Optimisation and Calculus of Variations*, 23(2):551–567, 2017. doi: 10.1051/cocv/2016003.

Filippo Santambrogio. *Optimal Transport for Applied Mathematicians: Calculus of Variations, PDEs, and Modeling*. Birkhäuser, 2015. doi: 10.1007/978-3-319-20828-2.

Isaac J. Schoenberg. Metric spaces and positive definite functions. *Transactions of the American Mathematical Society*, 44(3):522–536, 1938. doi: 10.1090/S0002-9947-1938-1501980-0.

Othmane Sebbouh, Marco Cuturi, and Gabriel Peyré. Structured transforms across spaces with cost-regularized optimal transport. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pages 586–594. PMLR, 2024. URL https://proceedings.mlr.press/v238/sebbouh24a.html.

Thibault Séjourné, François-Xavier Vialard, and Gabriel Peyré. The unbalanced gromov–wasserstein distance: Conic formulation and relaxation. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://proceedings.neurips.cc/paper/2021/hash/4990974d150d0de5e6e15a1454fe6b0f-Abstract.html.

Lou van den Dries. *Tame Topology and O-Minimal Structures*, volume 248 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, 1998. doi: 10.1017/CBO9780511525919.

Titouan Vayer. *A Contribution to Optimal Transport on Incomparable Spaces*. PhD thesis, Université Bretagne Sud, Vannes, France, 2020. URL https://arxiv.org/abs/2011.04447.

Cédric Villani. *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer, 2009. doi: 10.1007/978-3-540-71050-9.